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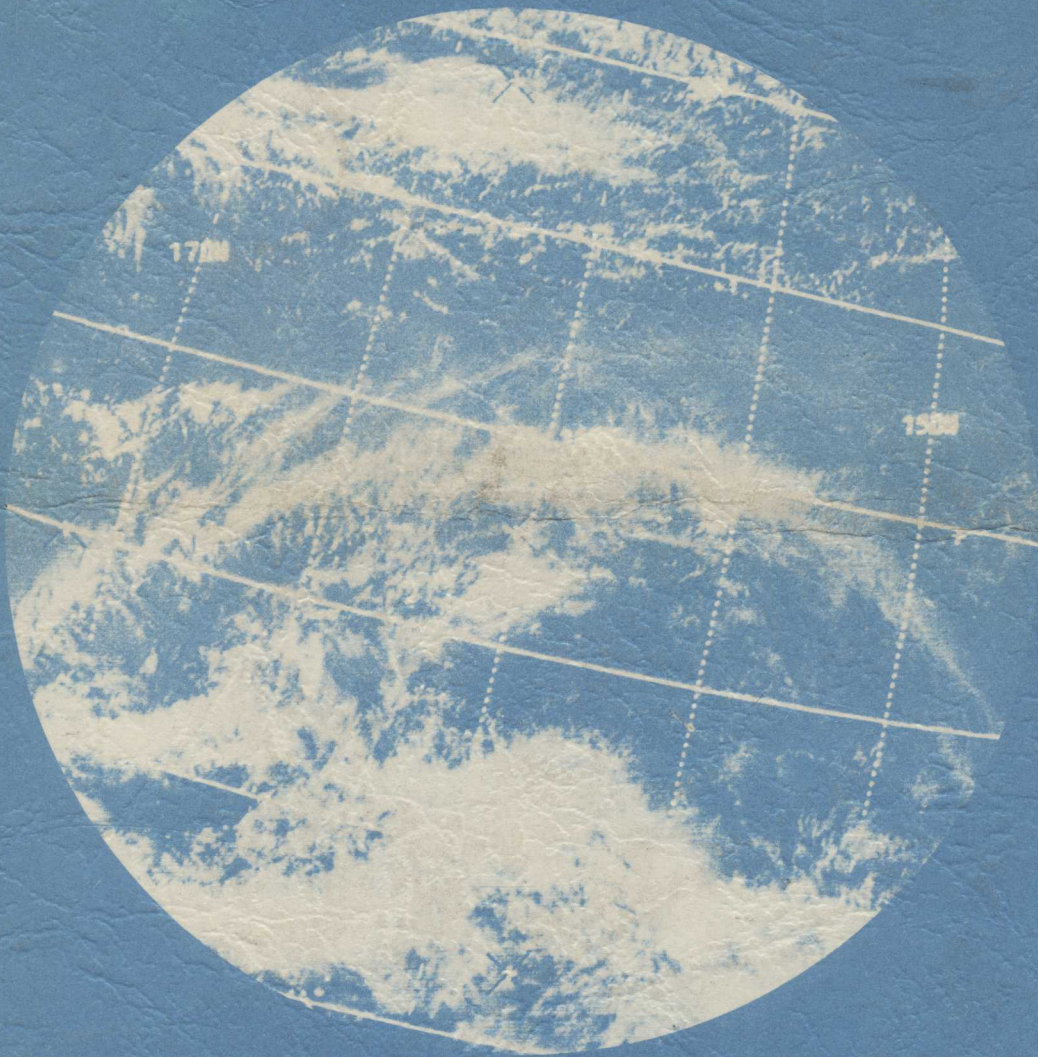
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THE MEAN WINDS OF THE UPPER TROPOSPHERE OVER THE CENTRAL AND EASTERN PACIFIC

by

JAMES C. SADLER
Department of Meteorology
University of Hawaii



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MONTEREY, CALIFORNIA 93940

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1. INTRODUCTION

The use of electronic navigation equipment has improved the general quality of aircraft wind observations (PIREP winds). The bulk of the observations for the past decade has been concentrated at the jet aircraft flying levels between 30,000 and 40,000 ft. These data have been used in calculating mean-monthly flow over tropical regions where data from fixed stations are extremely sparse. Pearson (1968) has used two years (1962 and 1963) of data from transequatorial flights to produce mean-monthly wind analyses over portions of the central and eastern Pacific. Additionally, averaged aircraft wind reports have been used in preparing the International Indian Ocean Expedition Meteorological Atlas (Ramage and Raman, 1972). In the Atlantic averaged PIREPS have been used in monthly streamline analyses (Aspliden, et al., 1966).

This atlas utilizes the PIREP winds from the first nine years of the commercial jet aircraft age in conjunction with radiowinds and radarwinds (rawins) from fixed stations to define the upper-troposphere wind climatology of the central and eastern Pacific Ocean.

2. DATA

2.1 Winds from PIREPS

Aircraft wind reports at 28,000 ft and above, for the area 10S to 50N between 155E and 120W for the period 1960-1968 were extracted from the manuscript maps of the National Weather Service at Honolulu Airport. These were supplemented east of 120W with data extracted from flight logs of the Canadian Pacific and UTA airlines. Monthly mean resultant winds were computed for each 5 degree latitude-longitude grid square and located at the averaged position of the reports. Wind steadiness* and percentage of observations with an easterly wind component were computed to aid in analysis and interpretation. The data are homogeneous in neither time nor space. With

each succeeding year the number of observations increased as did the average flight level. Observations are concentrated along heavily traveled air routes. The corridor between Honolulu and the west coast of the United States has about triple the data volume of any other route. The monthly data sample averages near 75,000 observations. A typical monthly distribution of data is shown in Figure 1.

Four sets of mean resultant winds were determined for each month. All altitudes were first composited. The observations were then stratified by height intervals to mesh with the rawins at mandatory constant pressure surfaces. The stratification intervals were: 28,000-32,000 ft (300 mb); 33,000-36,000 ft (250 mb); and 37,000 ft and above (200 mb).

2.2 Rawins

Updated summaries of wind data were prepared for most fixed land and ship stations over the ocean area, Mexico, Central America, and along the west coasts of South and North America. Where possible, the summaries covered the same period as the PIREP data. Data for stations not listed in "Monthly Climatic Data for the World" were obtained from various other sources. Data for most Southern Hemisphere stations, including the newly established French stations in the Society, Tuamotu, and Marquesas Islands, were extracted from the Daily Weather Bulletins of the New Zealand Meteorological Service. Summaries for the Pacific islands of Eniwetok, Kwajalein, Canton, and Midway, the islands of the Caribbean, and all stations in the United States except those along the west coast were obtained from the National Climatic Center (Asheville, North Carolina) and do not correspond in time to the period of the PIREPS. They are for varied lengths of record during the period 1945-1962.

$$\text{Wind steadiness} = \frac{\text{mean resultant wind speed}}{\text{mean wind speed}} \times 100\%.$$

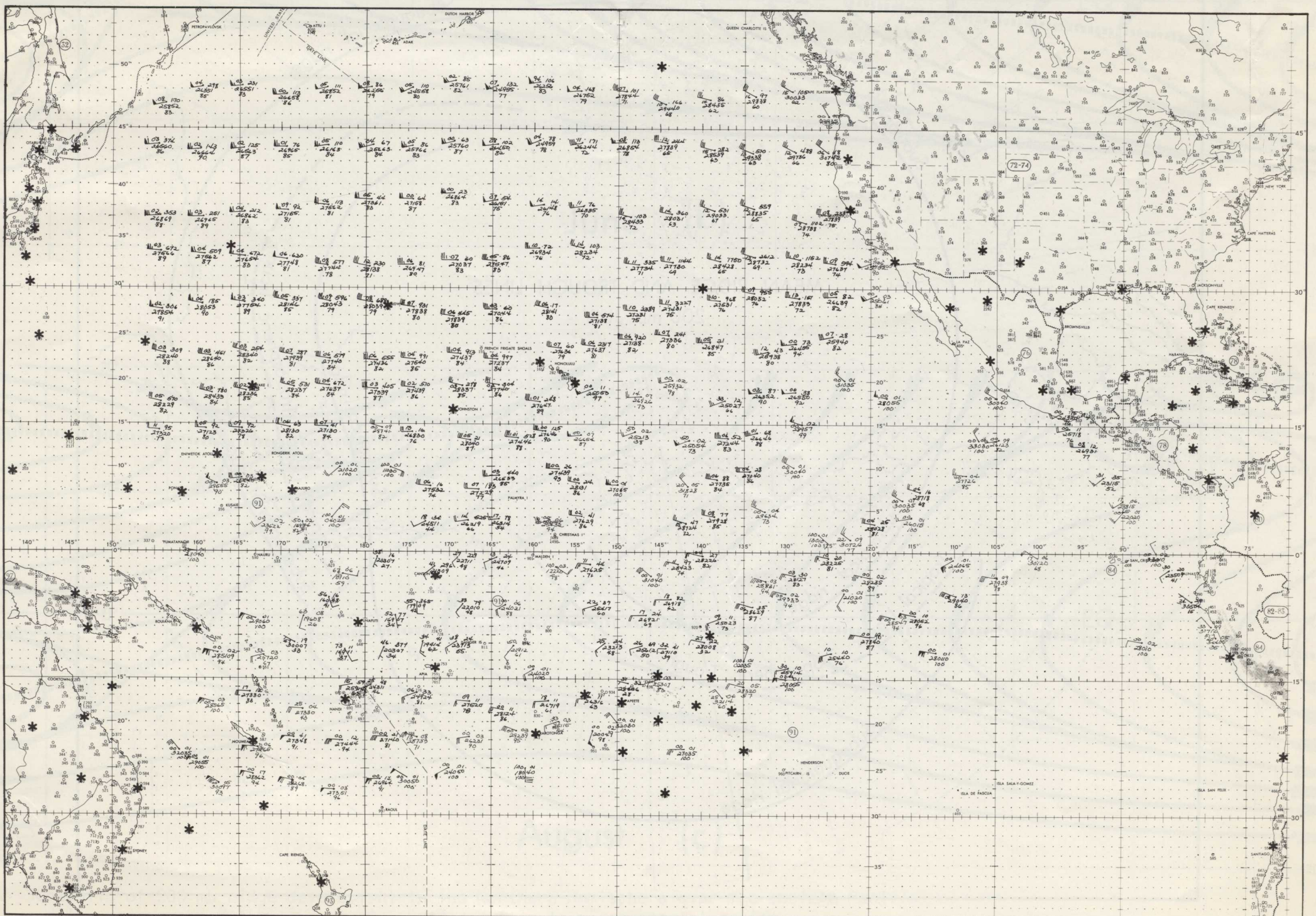
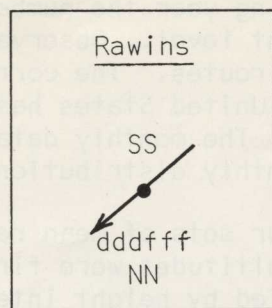
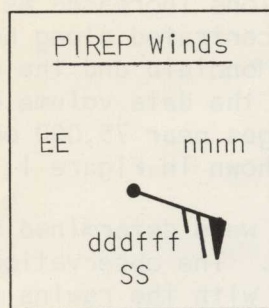


Figure 1. Typical monthly composite of aircraft reports. Asterisk symbol denotes location of rawin stations.

2.3 Plotting Models



- EE -percentage of winds with an east component
 nnnn -number of observations
 ddd -mean resultant wind direction
 fff -mean resultant wind speed in knots (flag = 50 knots, long barb = 10 knots, short barb = 5 knots)
 SS -steadiness of winds in percent
 NN -number of years of record

3. TERMINOLOGY

3.1 Systems

Since flow patterns are rather complex during the Northern Hemisphere summer, discussion is aided if major circulation features

are labelled. In Figure 2, the ridge system has a spiral configuration and double ridges exist over the oceans. When only a single ridge exists in a hemisphere or portions of a hemisphere, it is referred to as the subtropical ridge. When a double ridge system exists, the equatorward ridge will be referred to as the subequatorial ridge and the poleward one as the subtropical ridge. Note that the subequatorial ridge of the western and central North Pacific becomes the subtropical ridge across the United States and the North Atlantic. The subequatorial ridge of the eastern North Pacific, northern Africa, Asia, and the North Pacific.

The trough between the subequatorial and subtropical ridges will, in general, be referred to as the Tropical Upper-Tropospheric Trough (TUTT). It is commonly called the Mid-Pacific Trough (MPT) in the central Pacific and the Mid-Atlantic Trough (MAT) in the Atlantic. The term "Tropical Upper-Tropospheric Trough" is preferred since similar troughs exist during summer in the Southern Hemisphere. In the eastern North Pacific, Central America, and the Gulf of Mexico on occasion for shorter periods it is observed as a continuous feature extending westward from the Mediterranean Sea to Asia (Sadler,

Another near-equatorial system which needs defining is the Tropical Upper-Tropospheric Trough (TUTT). It is commonly called the Mid-Pacific Trough (MPT) in the central Pacific and the Mid-Atlantic Trough (MAT) in the Atlantic. The term "Tropical Upper-Tropospheric Trough" is preferred since similar troughs exist during summer in the Southern Hemisphere. In the eastern North Pacific, Central America, and the Gulf of Mexico on occasion for shorter periods it is observed as a continuous feature extending westward from the Mediterranean Sea to Asia (Sadler,

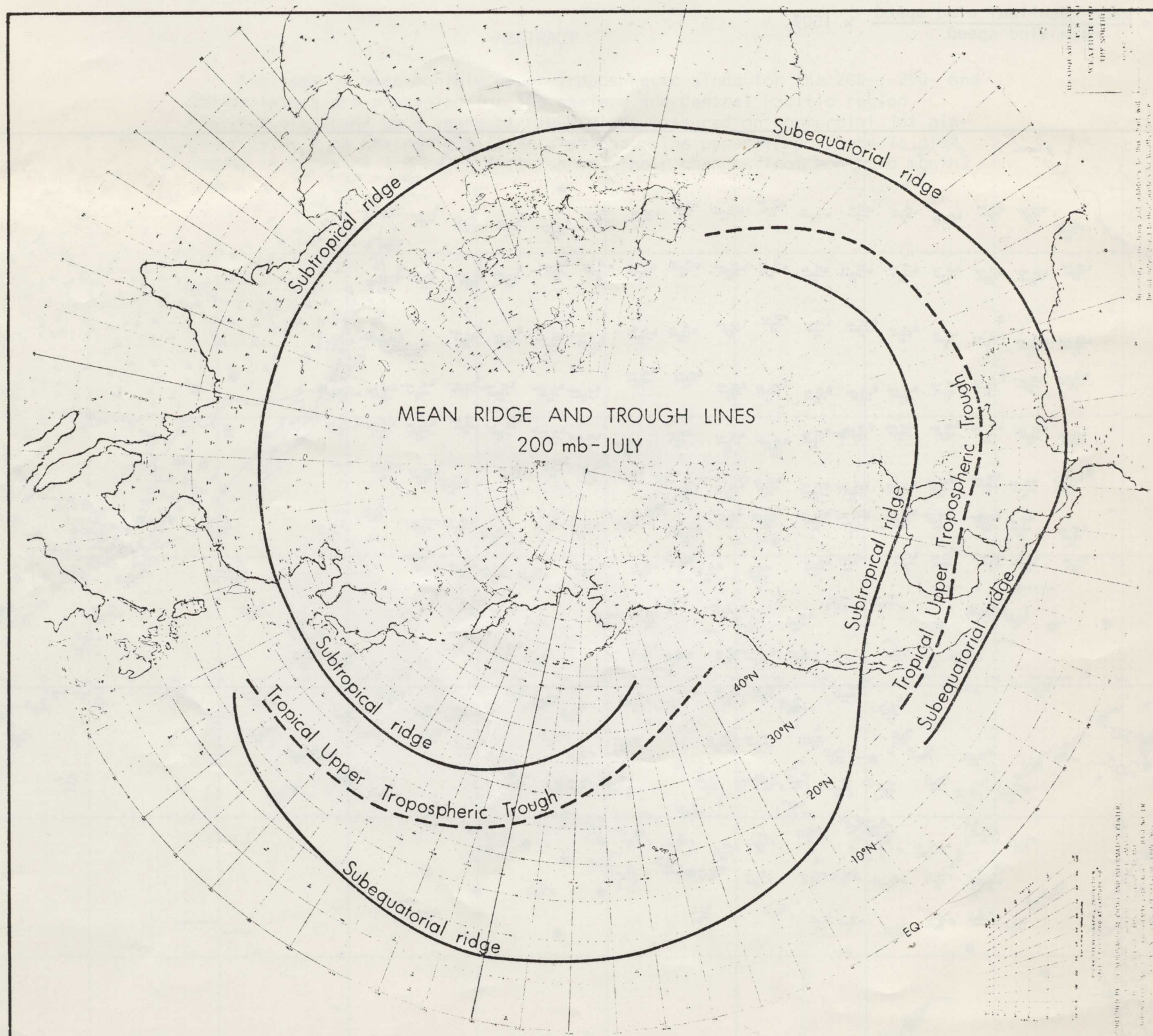
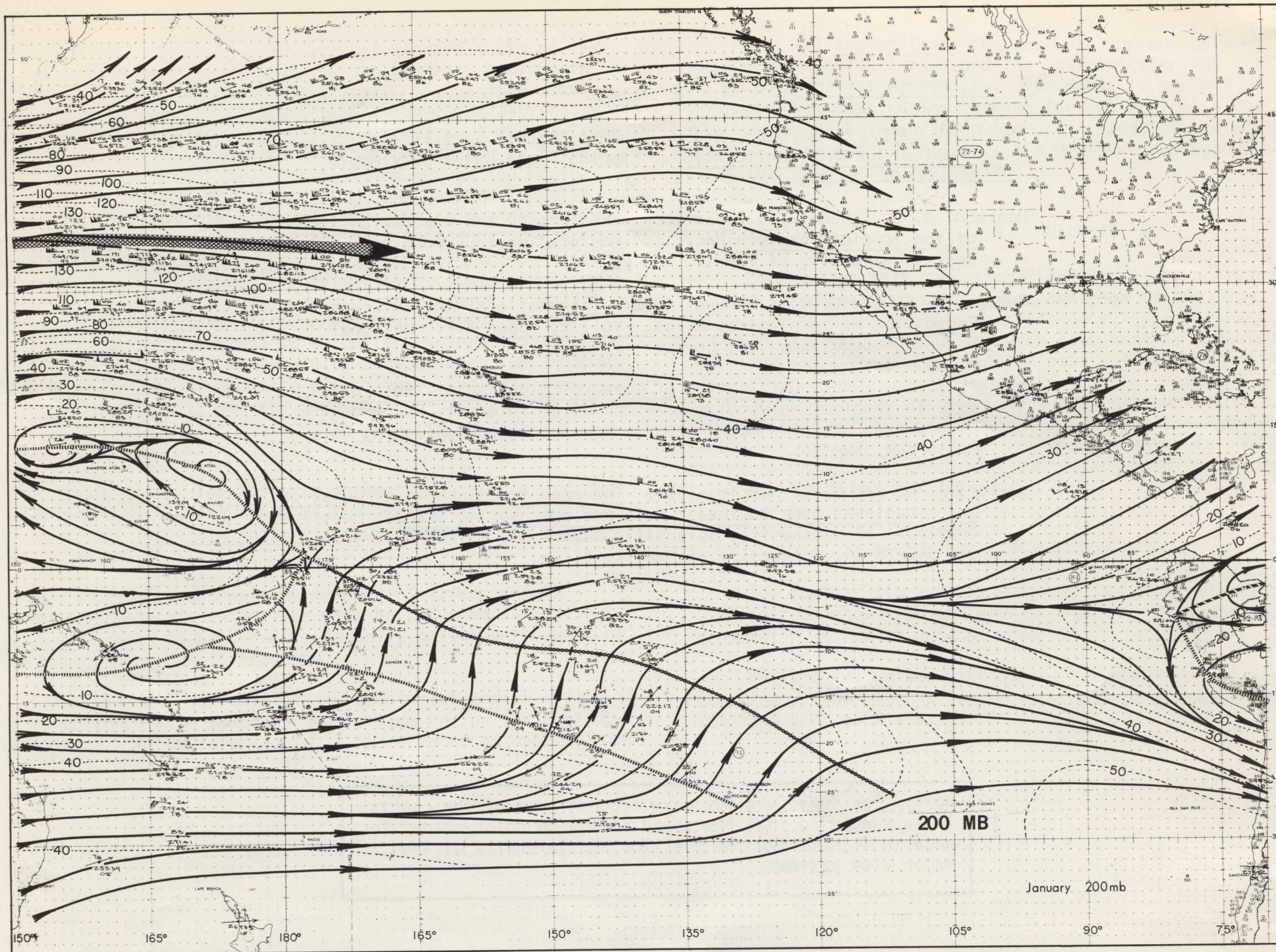


Figure 2. Mean July positions of ridges and troughs at 200 mb.



JANUARY

Northern Hemisphere

A steady, persistent temperate jet stream with a core speed of near 135 kt enters the area along 35N. Speed decreases rather rapidly eastward, and there is a "fanning" in direction downstream from the core center with an apparent splitting into two branches. The northernmost branch enters the United States near 46N as an anticyclonic turning current of near 50 kt maximum speed. The southern branch passes just north of Hawaii and enters the American coast over Baja California as a cyclonic turning current. The maximum speed decreases to less than 50 kt east of Hawaii and increases again to 50 kt on the west coast. Between the two branches there is an area of minimum speed over the eastern Pacific.

Note that no winds with an easterly component are observed in the high steadiness (95 + %) region of the temperate jet west of 165W. This indicates that no deep westerly troughs with cut-off lows occur in this region as opposed to the eastern North Pacific which is a region of cut-off lows and blocking highs with omega patterns.

The subtropical ridge, near 14N at 155E, terminates near 177W, and westerly flow covers the tropical region eastward of 177W through Central America and northwestern South America.

Southern Hemisphere

The subtropical ridge, near 12S at 155E, branches near 170E with the equatorward branch terminating near 177W at the juncture with the northern hemisphere subtropical ridge and the poleward branch extending east-southeast some 10 degrees south of and paralleling the tropical upper tropospheric trough (TUTT).

The core of temperate westerlies with a maximum speed of approximately 45 kt lies near 25S over eastern Australia. Speeds decrease eastward to the subtropical ridge--TUTT systems between 110W and 130W, and then increase to greater than 50 kt over western South America where the northwesterly stream of subtropical westerlies from the equatorial region merges with the temperate westerlies.

Equatorial Region

Equatorial easterlies exist from near 177W westward, between the two subtropical ridges. Westerlies cover the equatorial region from near 177W eastward to South America.

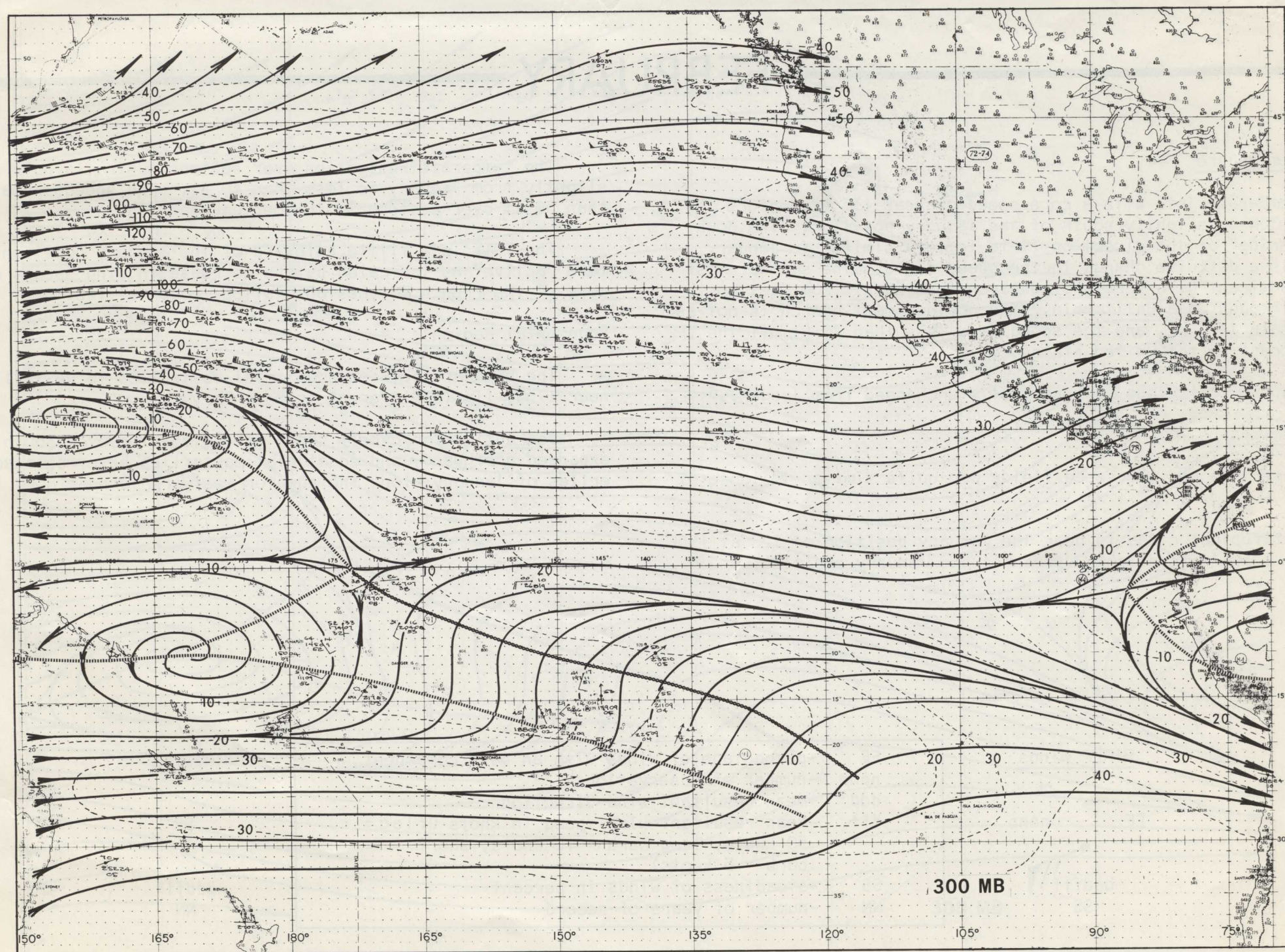
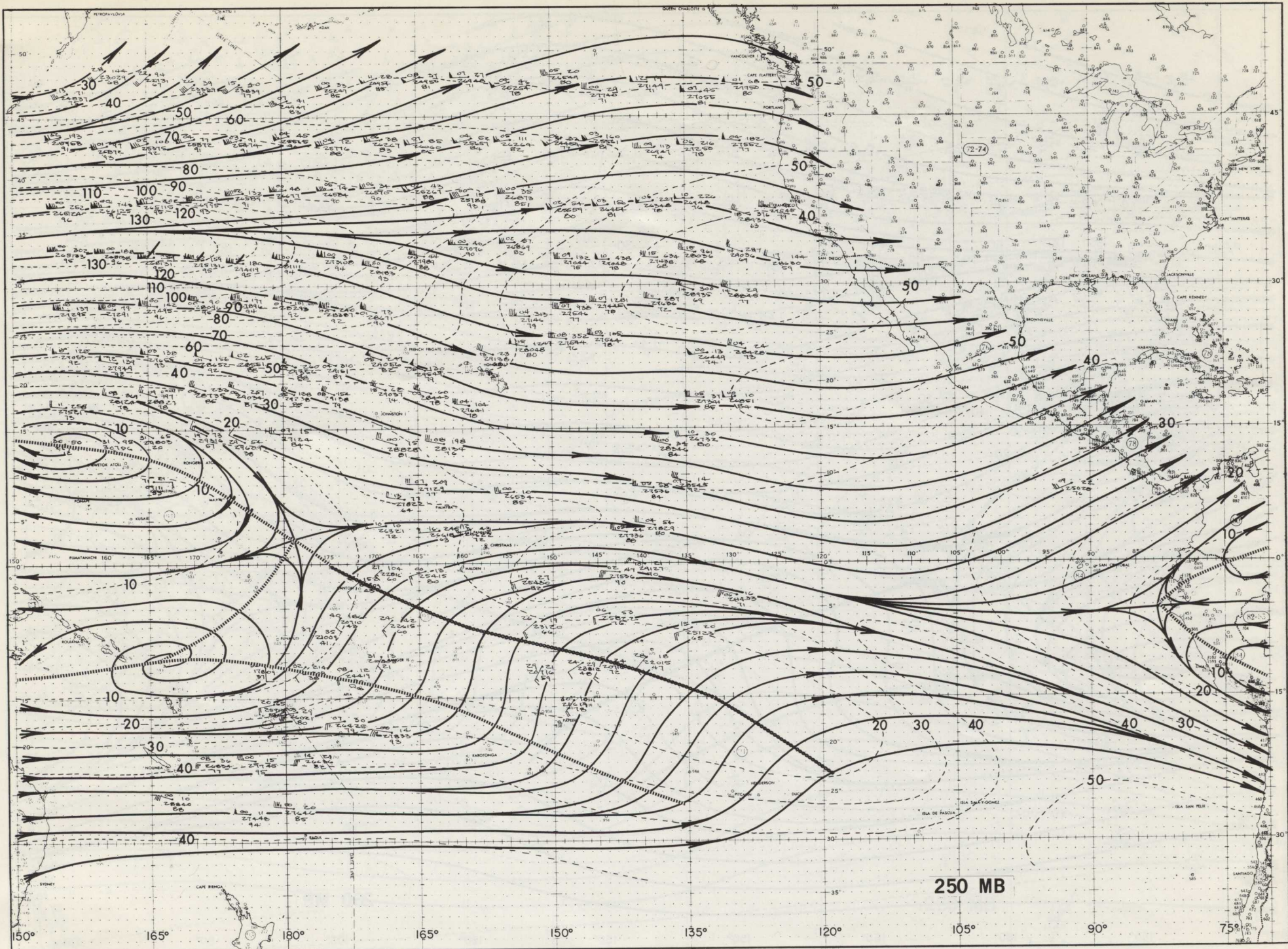
PIREP Winds

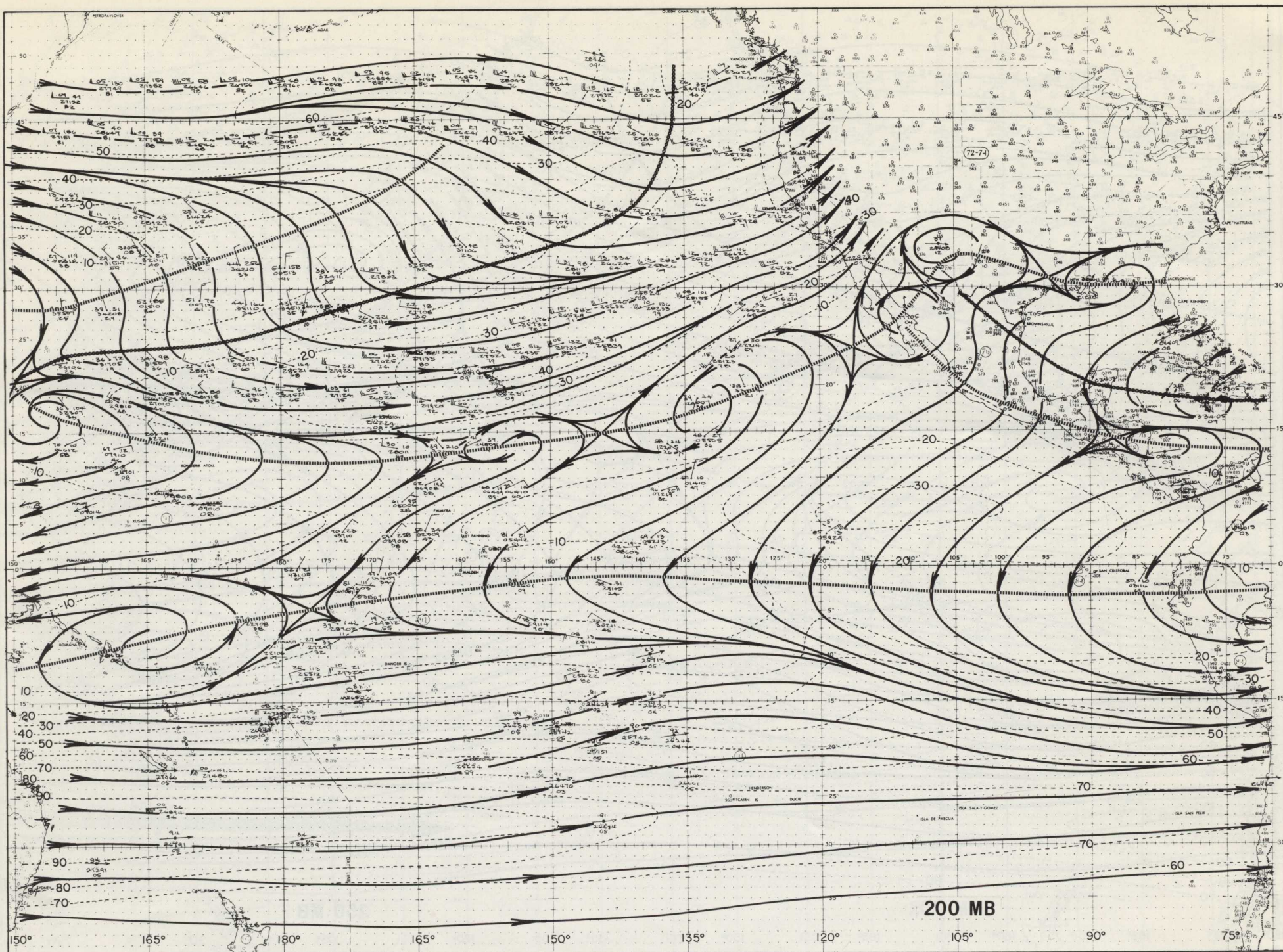
EE nnnn
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Rawins

SS
 dddfff
 NN





AUGUST

Northern Hemisphere

All systems have moved northward. The subequatorial ridge has intensified and is continuous from the western Pacific to South America. The circulation patterns and major current speeds have changed little from July.

Southern Hemisphere

The subtropical ridge is near 10°S at 155°E and links with the buffer system which lies south of the equator across the entire eastern Pacific.

The temperate jet position and intensity have changed little from July. The trough in the westerlies has moved eastward of 135°W.

Equatorial Region

This is the first month in the year that easterly winds are found on the equator at all longitudes. The speed of the northeast current outflowing from the eastern Pacific low-level monsoon has increased to near 30 kt.

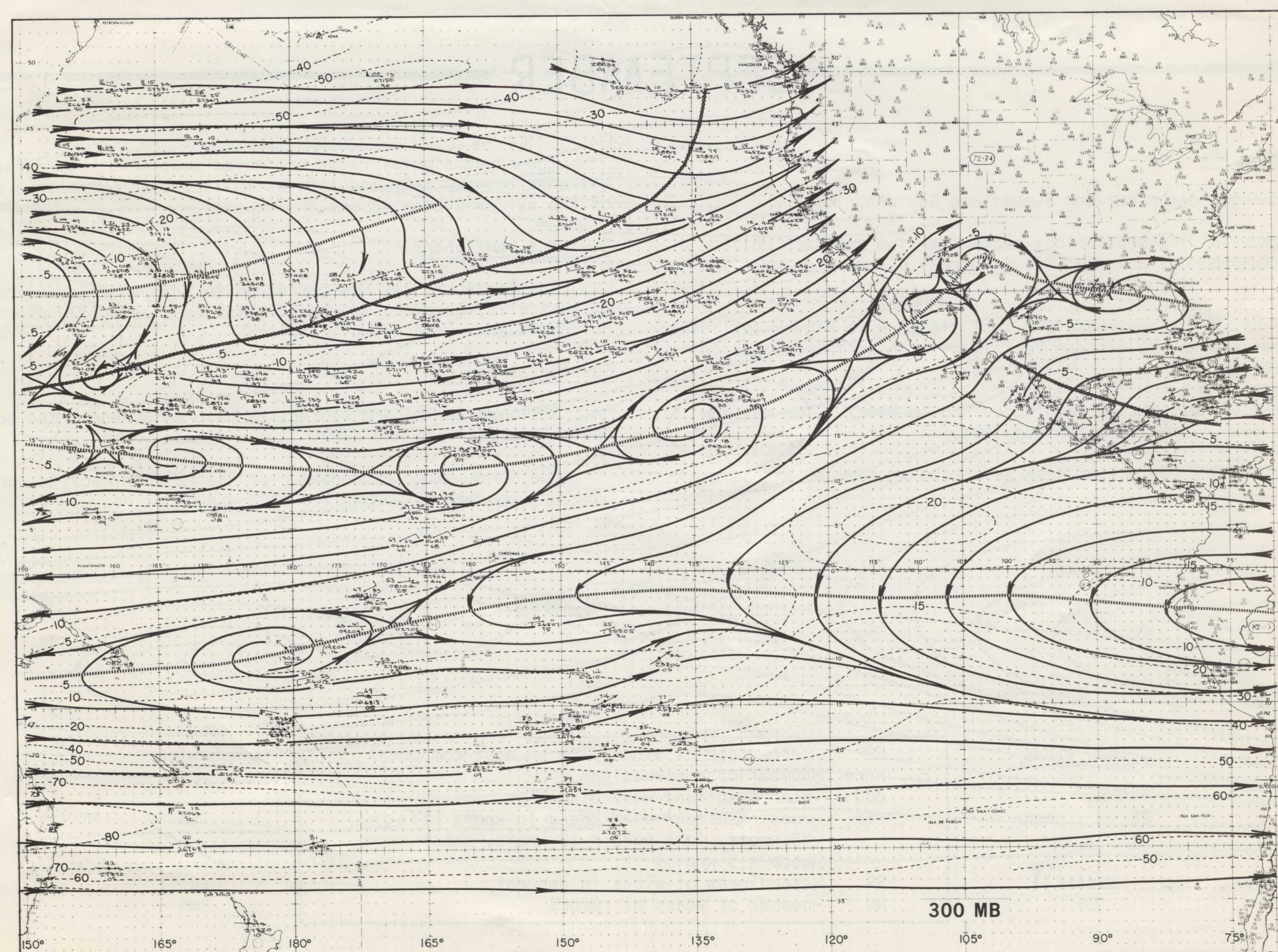
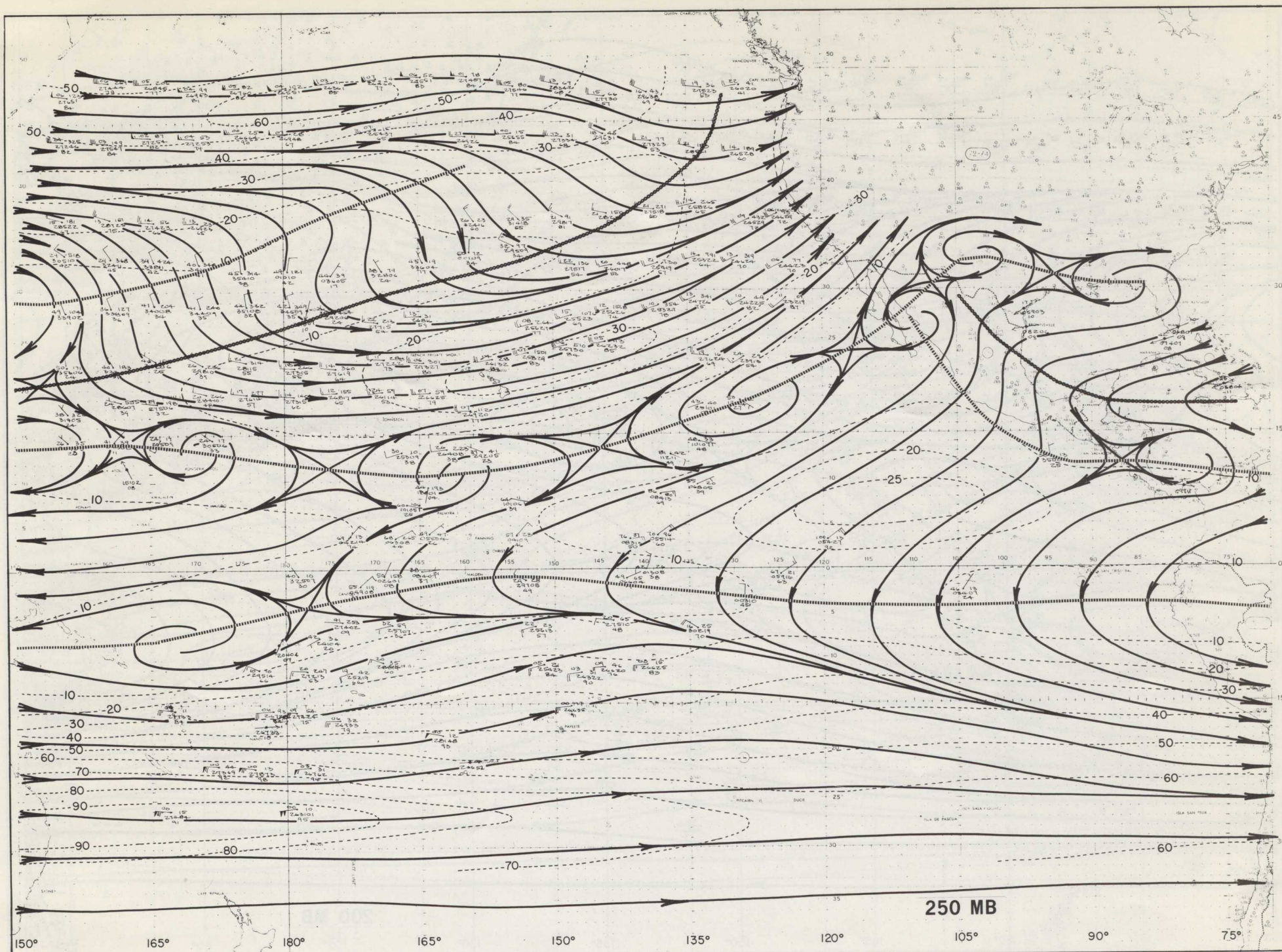
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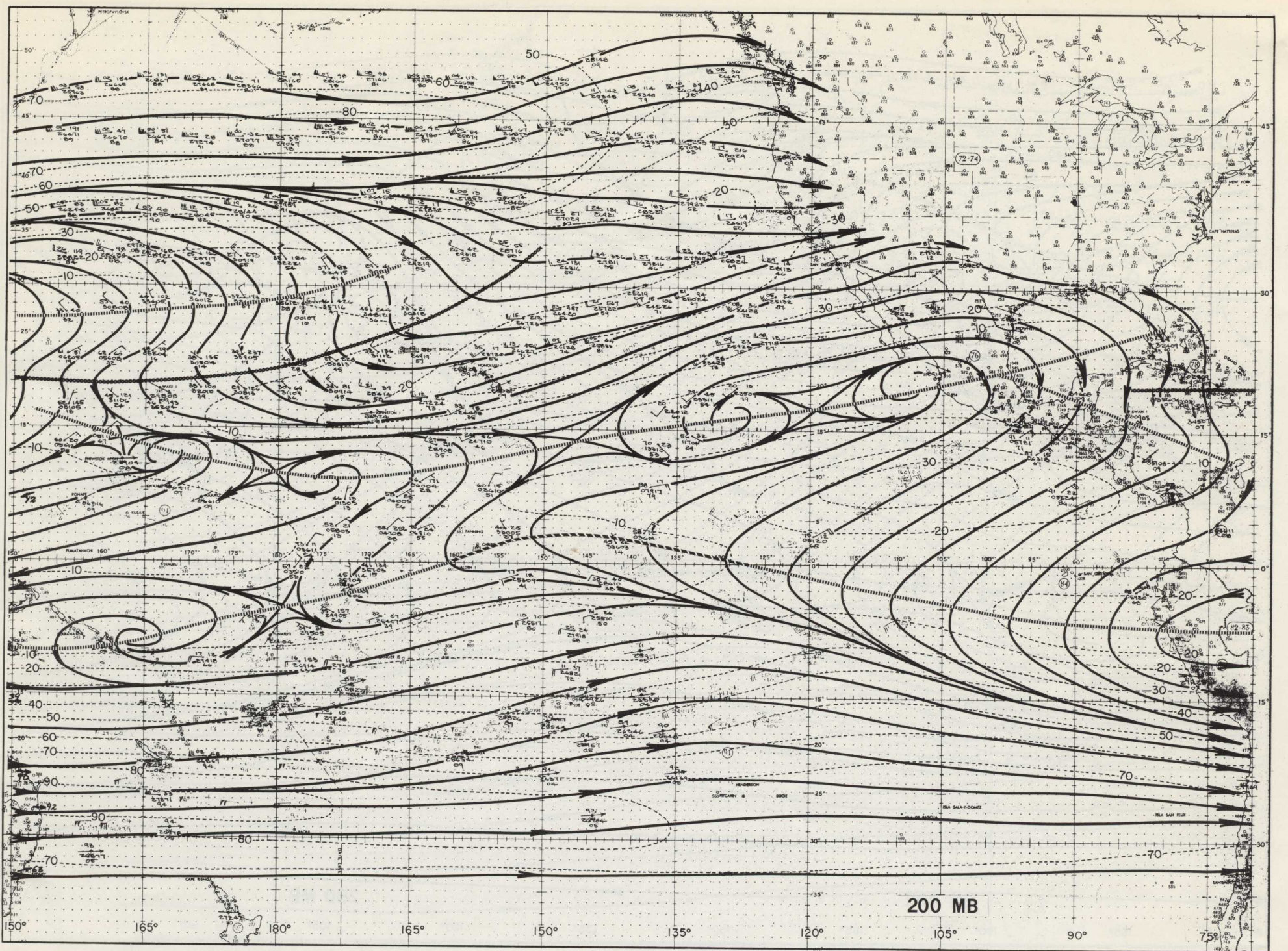
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 NN





SEPTEMBER

Northern Hemisphere

Large changes occur between August and September.

The segment of the subtropical ridge over the southern United States has disappeared, and the TUTT is barely discernible over the Caribbean.

The MPT has moved southward and lost its link to higher latitudes as ridging returns to the eastern Pacific off the northwest coast of the United States.

The temperate westerlies have increased to 80 kt. The maximum core remains near 45N between 180 and 165W.

The subtropical westerlies have decreased to near 30 kt eastward of Hawaii.

The subequatorial ridge has moved southward. It apparently has weakened between 165W and 130W as is indicated by northward movement of the buffer system across the equator.

Southern Hemisphere

The temperate jet has decreased to near 90 kt over and just east of Australia; however, speeds have increased slightly over French Polynesia and western South America.

Equatorial Region

Southern Hemisphere westerlies have returned to the equatorial region between 165W and 130W.

The northeasterly current over the eastern Pacific north of the equator remains well organized with speeds of near 30 kt.

PIREP Winds

EE nnnn

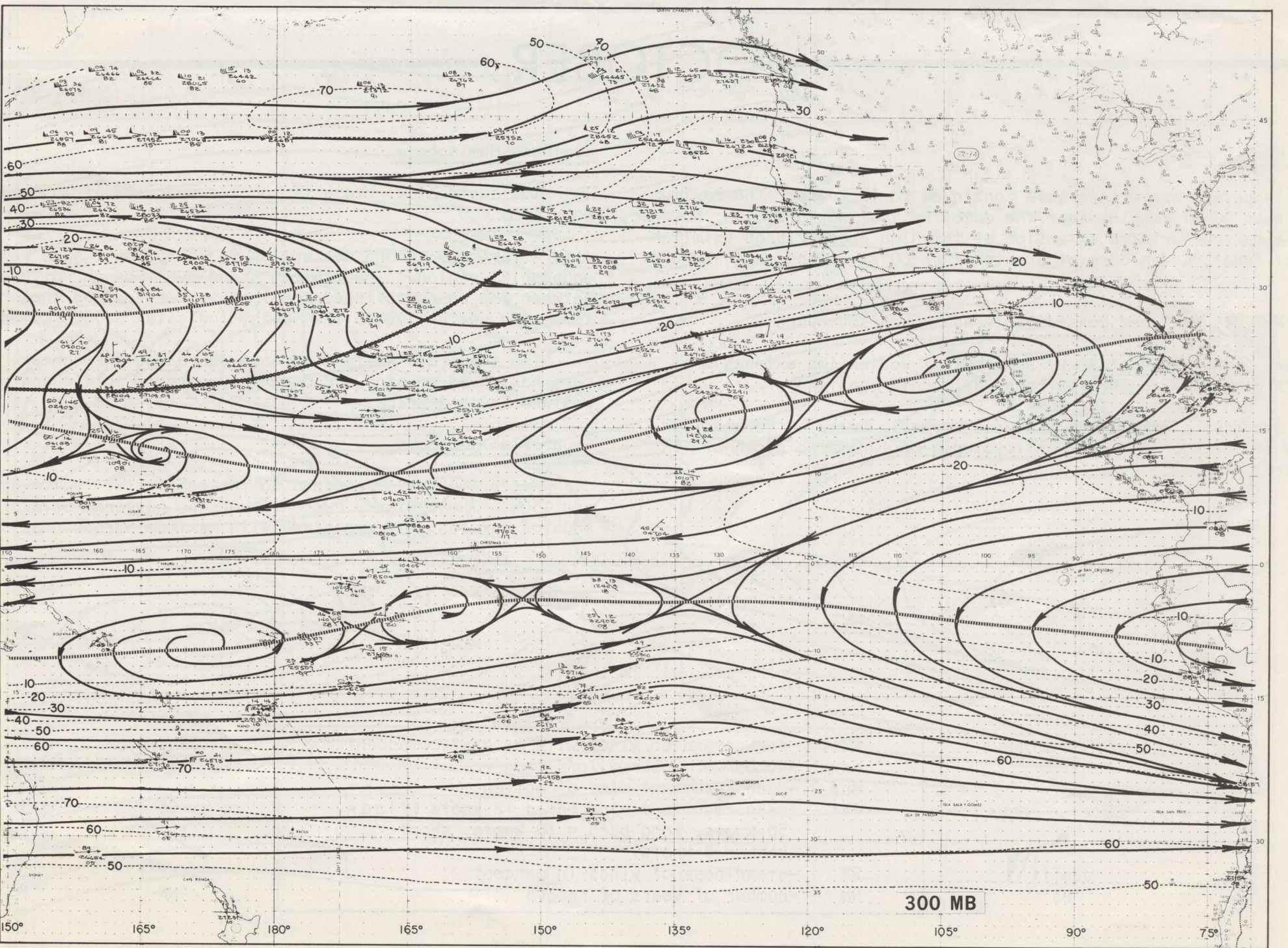
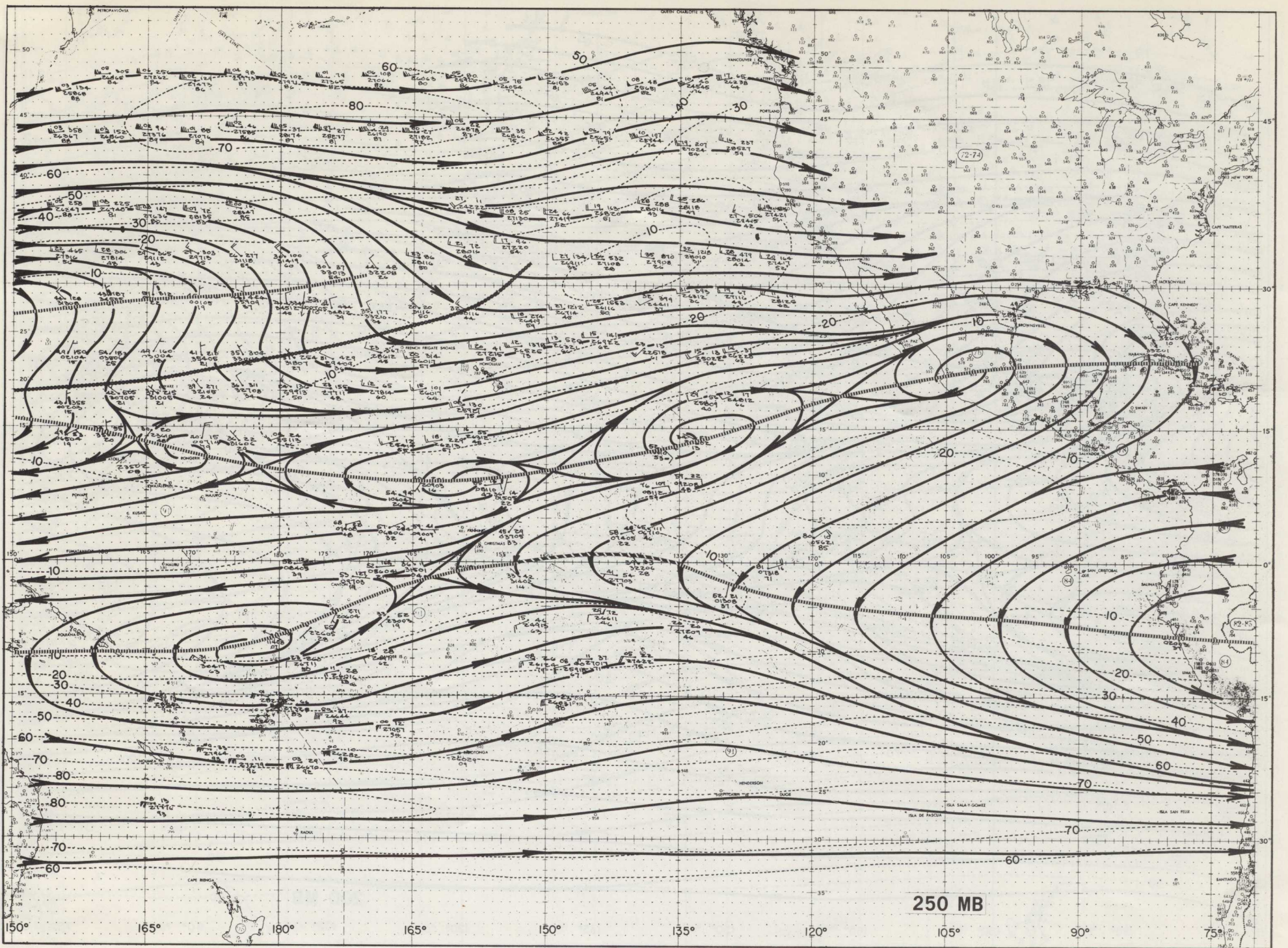
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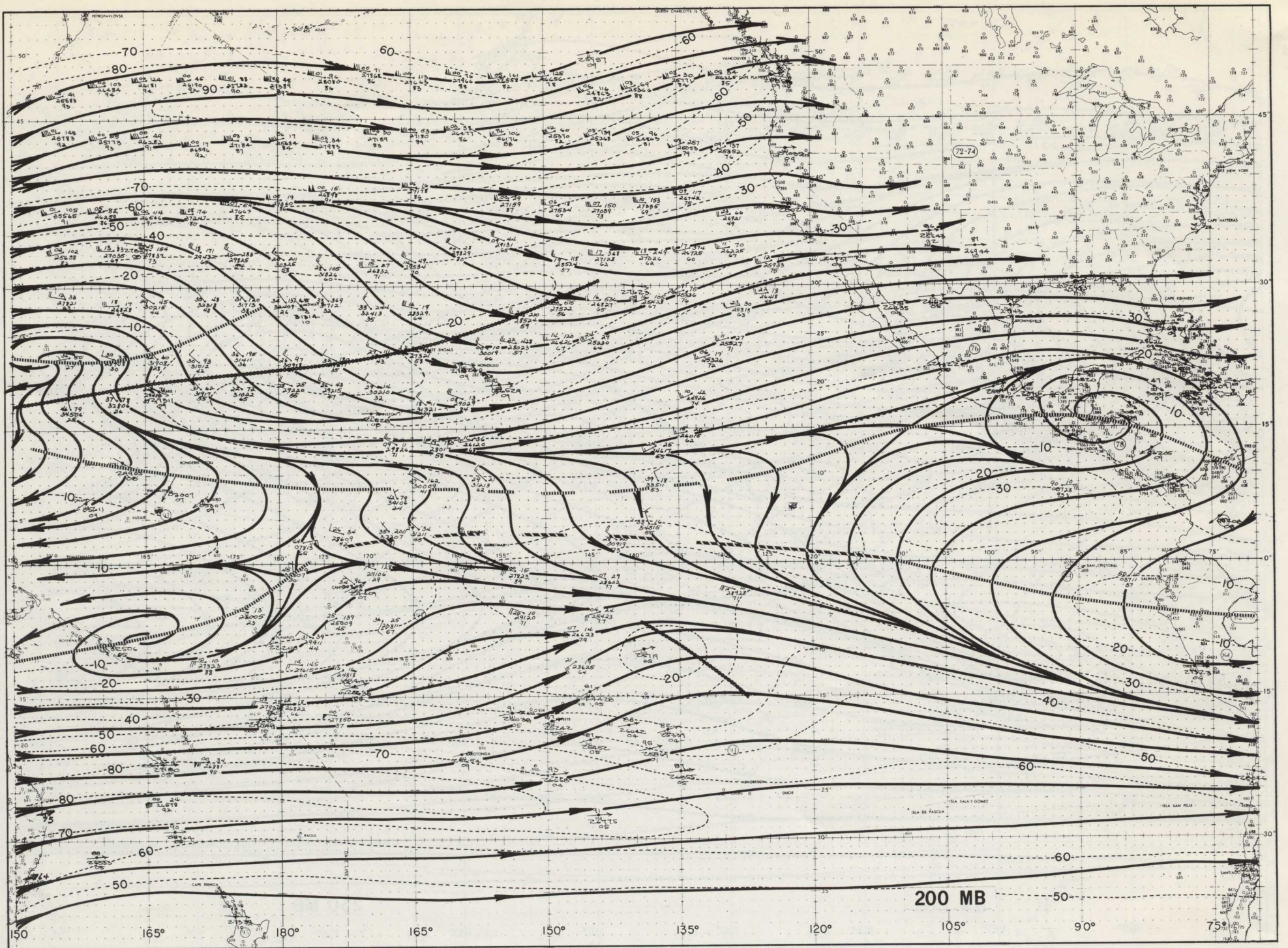
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Rawins

SS

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OCTOBER

Northern Hemisphere

The temperate westerlies have continued to increase. The maximum speed of near 90 kt remains near the dateline just north of 45N; however, there is a marked eastward extension of the high speed ribbon (similar to that of June), and a core of greater than 60 kt enters southwestern Canada.

The MPT has weakened considerably and moved southward.

A very thin ribbon of 30-kt subtropical westerlies extends from just south of Hawaii to the southern California coast.

The subtropical ridge has weakened markedly between 180 and 120W and moved southward. A strong anticyclonic cell remains north of 15N over Central America.

Southern Hemisphere

The subtropical ridge is located near 10S near 180.

Relatively strong westerlies of 25 to 30 kt are found in the equatorial region. The minimum speed over French Polynesia between the re-established tropical westerlies and the temperate westerlies to the south, indicates a weak ridge.

The temperate westerlies have decreased to 20 kt. The core has moved equatorward over the western Pacific and western South America.

Equatorial Region

Westerlies have returned to the equatorial region near 180W and 130W. The easterly current over the eastern Pacific remains well organized with maximum speeds near 30 kt.

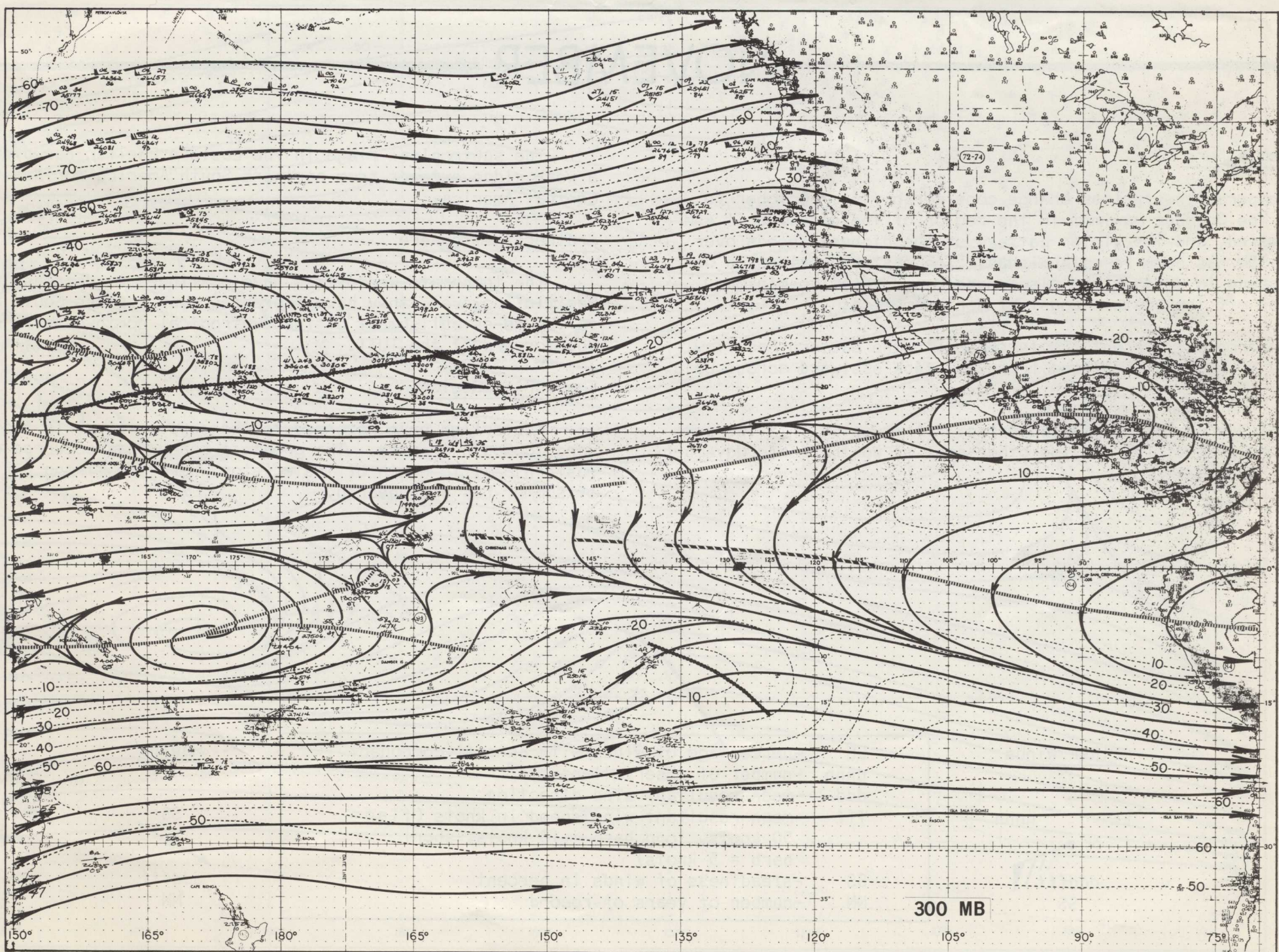
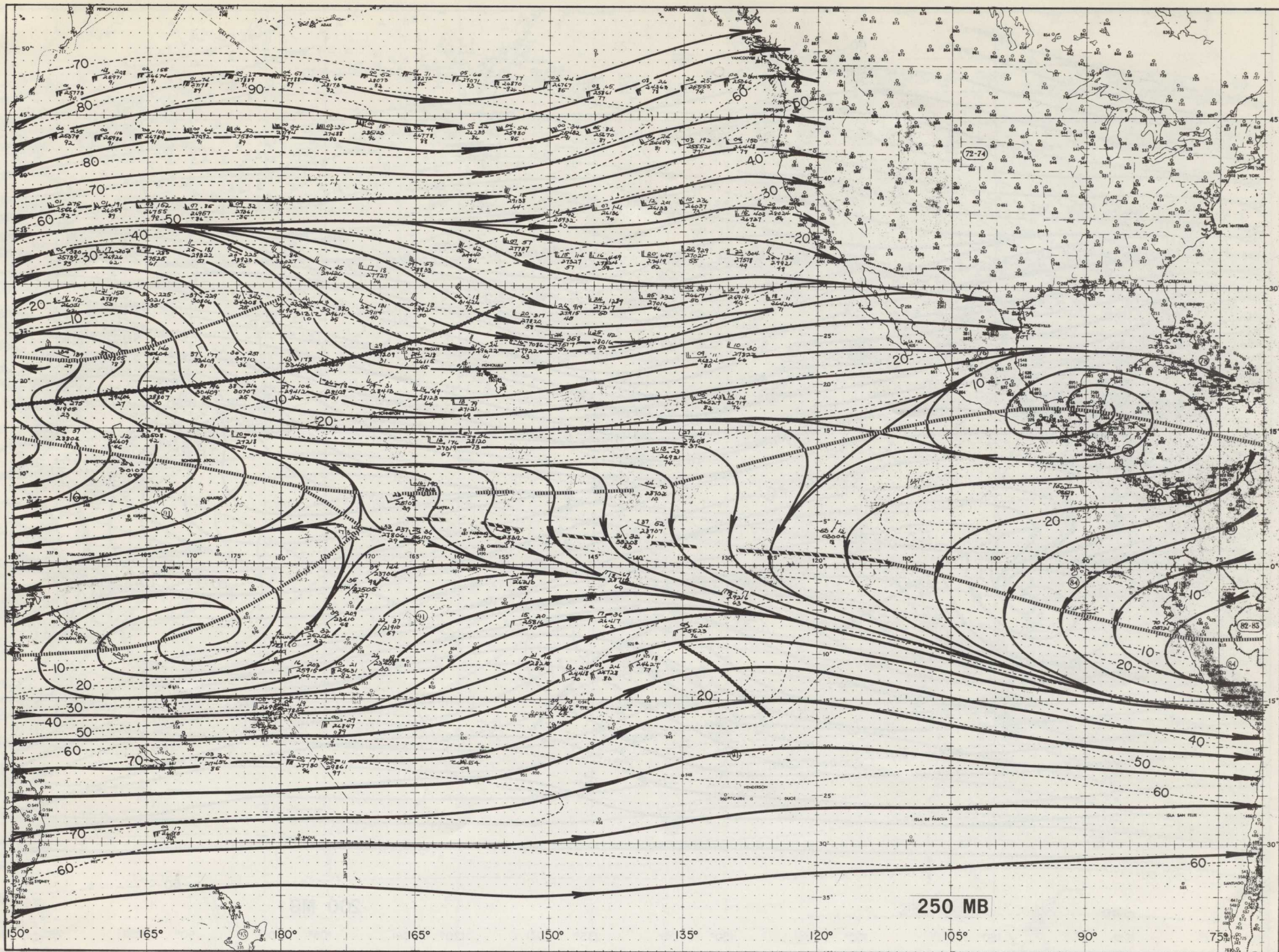
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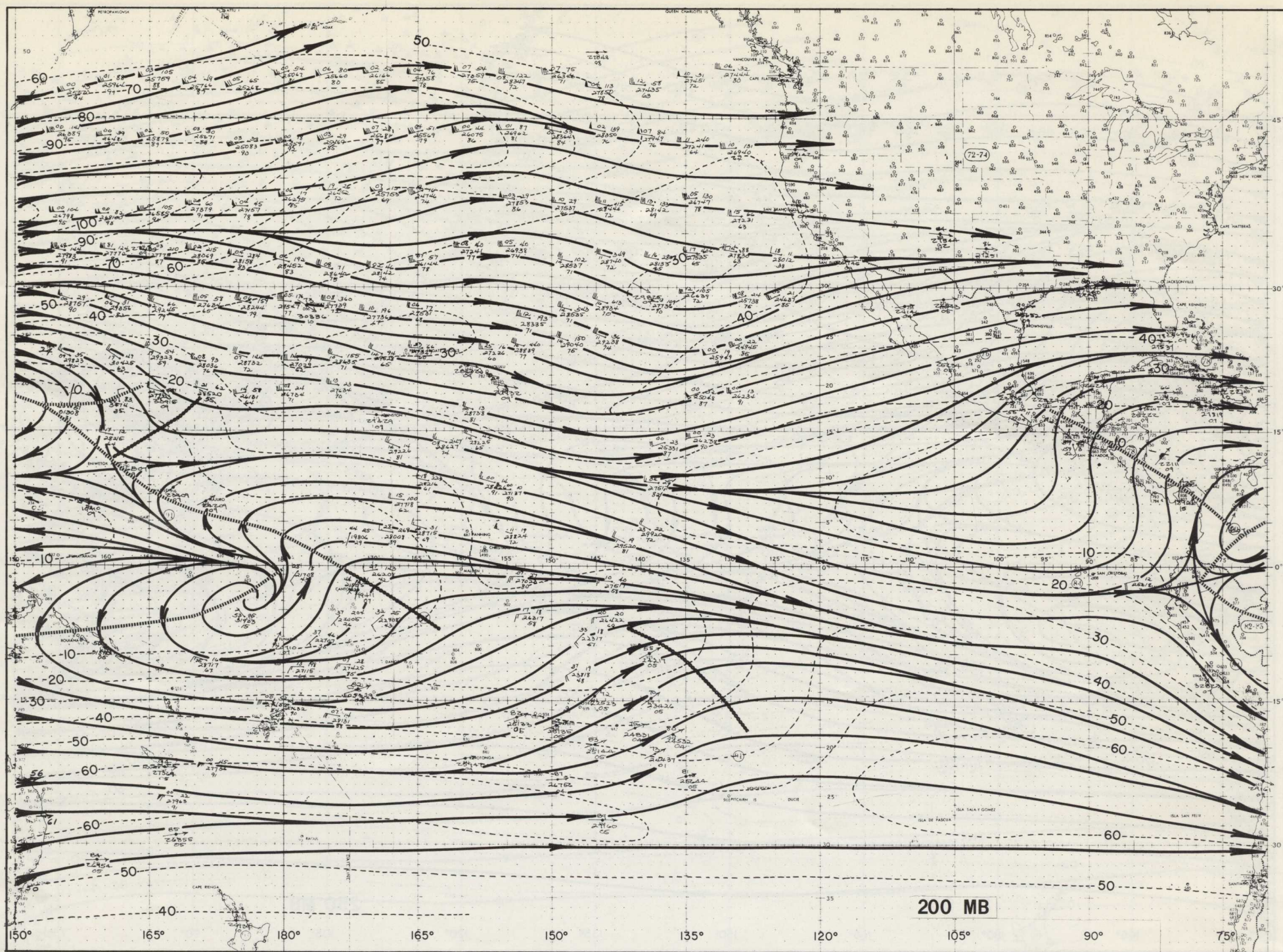
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Rawins

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NOVEMBER

Northern Hemisphere

The temperate westerlies change considerably from October to November. The October maximum in the region of the dateline at 45N has disappeared, and the winter pattern of the highest speeds over Japan near 35N has been re-established. The eastward extension of the high speeds in October has disappeared, and relative minimum speeds exist off the western coast of North America.

The subtropical current has increased in the eastern Pacific. Core speeds of near 45 kt extend from 22N, 135W eastward across Baja California.

The MPT has essentially disappeared. It is split into two weak segments, one west of 180 and the other east of weather ship November (30N, 140W) and Hawaii.

The subtropical ridge in the eastern Pacific has disappeared. A weak ridge extends from northern South America northwestward to Central America.

Southern Hemisphere

The temperate westerlies have continued to decrease in the eastern Pacific; however, they have increased over western South America to the re-establishment of the subtropical northwesterly system equatorward of the TUTT.

Equatorial Region

Westerlies have returned to the equatorial strip from the eastern Pacific north of the equator where the strong northwesterly of summer has disappeared.

PIREP Winds

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